

MONEY FOR MARS

FACILITATOR'S GUIDE

An exploration workshop for teams working at the edge of the known.

Money for Mars

A Space Economy Design Game

What happens to money when a message takes hours to arrive?

How do you buy something when the seller is on Mars?

Who do you trust to hold or exchange value on a new frontier?

Facilitator Guide

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Created by Scott Smith and Susan Cox-Smith, Changeist

moneyformars.com · changeist.com · github.com/Changeist/money-for-mars

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About This Guide

This guide tells you how to run Money for Mars. The cards themselves — locations, personas, financial tools, and crisis events — live in a separate spreadsheet, the Card Ledger. The ledger is where you build a deck for your specific session. This guide tells you how to play once you have one.

The split exists for a reason. Workshop guides that bundle their content into the document have to bloat to add new content, and any content not in the document reads as second-tier. Keeping the rules and the cards in separate files lets each grow on its own terms.

What you need to run a session

- This guide.
- A printed deck of cards. The Core set in the Card Ledger is the general deck and needs no assembly. The five audience tabs are an extended menu to add from or swap in. “Producing the deck” below covers what goes on a card face.
- A canvas worksheet for each team (template in the appendix).
- Wall space, large paper, or a digital whiteboard.
- A timer everyone can see.

Expert sessions

Contact ssmith@changeist.com if you would like to discuss running an expert-led Money for Mars workshop for your team or organization. This format can also be adapted for custom sessions that extend the exploration.

What This Game Is

Most organizations already have the ingredients for innovation — familiar roles, known tools, existing systems. What they lack is the right conditions to question them.

Money for Mars puts teams in an environment where the assumptions behind those tools break down. Teams draw three cards — a place in the solar system, a person who needs financial services, and a financial tool from Earth — and invent something that would actually work where Earth's version wouldn't. Space is the prompt. The thinking it produces comes home.

The exercise is not about predicting space economies. It is about developing the muscle for questioning what any system takes for granted — and redesigning it when those assumptions no longer hold. That muscle transfers. Frontier markets, emerging technology, regulatory disruption, post-disruption strategy — anywhere the old tools don't quite reach.

Why this works

Take any tool — financial, organizational, commercial — and ask: what does it assume about the world? Fast communication. Enforceable contracts. Shared time. Recoverable counterparties. Legal jurisdiction. Trust at a distance. Strip those assumptions out — as space requires — and you discover what the tool actually does, what it depends on, and how it might be rebuilt.

The friction is the point. Designing through unfamiliar constraints builds exactly the cognitive capacity that transfers to any frontier context. Participants don't need to know about space finance to engage. They need to be willing to ask why something works the way it does — and the exercise makes that question unavoidable.

What this builds

Four things participants consistently leave with, across audience types:

- **Assumption fluency.** The ability to name what a system takes for granted and redesign it when those assumptions change.
- **Creative perspective-taking.** Inhabiting unfamiliar roles and contexts surfaces blind spots that familiar framing conceals.
- **Collaborative problem-solving.** Engineering, commercial, strategy, and policy people work the same design challenge. The exercise requires all of them; the language is accessible to all of them.
- **Commercial horizon-scanning.** Space conditions are a proxy for real frontier problems. Participants regularly surface applicable near-term opportunities they hadn't seen before.

Choose Your Format

Four formats. Pick the one that matches your time and audience.

Teams are two to five people. Below two there is nobody to argue with; above five somebody stops contributing. Every schedule below allows four minutes per team for sharing, which includes the handover. That is the number to recalculate from if your team count changes.

Quick Play — 90 minutes

Best for: online sessions, first-time players, recorded sessions, conference slots. Teams design concepts and share them. No ecosystem phase. Recommended team count: 4–6, so 8 to 30 people.

Segment	Time
Introduction and card draw	10 min
Design work (teams)	40 min
Sharing concepts	25 min
Debrief	15 min

Note on sharing time: 25 minutes covers six teams at four minutes each. With four teams you get nine minutes back — put them into design work rather than into the debrief, which is already sized.

Quick Play + Crisis — 2 hours

Best for: groups interested in resilience and stress-testing. Quick Play, then a Crisis Card that every team has to survive. Recommended team count: 4–6, so 8 to 30 people.

Segment	Time
Introduction and card draw	10 min
Design work (teams)	40 min
Sharing concepts	25 min
Crisis Card drawn and read	5 min
Adapting the design (teams)	10 min
Crisis share-back	15 min
Debrief	15 min

Full Game — 3.5 hours

Best for: in-person workshops, team off-sites, longer training. Both phases: design concepts, then connect them into ecosystems. Includes a 15-minute break. Recommended team count: 6–9, so 12 to 45 people.

Segment	Time
Introduction and card draw	15 min
Phase 1 — design (teams)	45 min
Phase 1 — connection tags	5 min
Phase 1 — sharing	40 min
Break	15 min
Phase 2 — forming clusters	10 min
Phase 2 — mapping the cluster	15 min
Phase 2 — designing the shared institution	30 min
Phase 2 — cluster share-back	20 min
Debrief	15 min

Why 3.5 hours and not 3: nine teams sharing at four minutes each is 36 minutes, and Phase 2 has four distinct steps rather than the two the schedule used to name. The old three-hour version fitted by squeezing the debrief and by giving cluster formation and cluster mapping the same fifteen minutes. This is the honest length. If you only have three hours, run Quick Play + Crisis instead of a compressed Full Game.

Two Sessions — 90 × 2

Best for: university courses, online cohorts, distributed teams across time zones. Session 1 is design and share. Session 2 is ecosystem formation. The facilitator does between-session work to prepare clusters. Recommended team count: 4–6, so 8 to 30 people.

Session 1

Segment	Time
Introduction and card draw	15 min
Phase 1 — design (teams)	45 min
Phase 1 — connection tags	5 min
Phase 1 — sharing	25 min

Session 2

Segment	Time
Recap and cluster briefing	10 min
Mapping the cluster	15 min
Designing the shared institution	30 min
Cluster share-back	20 min
Debrief	15 min

Critical: over-recruit by 20–30%. Two-session formats lose participants to scheduling, travel, and competing demands. Session 2 will not have everyone who attended Session 1.

Between sessions, the facilitator collects each team's canvas, identifies 2–3 ecosystem clusters by location proximity or trade complementarity, and sends each participant a short pre-read 24 hours before Session 2 with their cluster mates' concepts.

The Cards

Each team draws three cards — one location, one persona, one financial tool. The job is to invent something that works for all three.

The full set of cards lives in the Card Ledger spreadsheet. This section explains what each deck is for. The ledger has the cards.

Only one piece of card information is a play variable. Location cards carry a zone — Near Earth, Solar System, or Deep Space. The zone sets communication latency, and latency sets everything the design has to answer: how long a message takes, how long help takes, whether a contract can be enforced, whose rules apply. A location card has to tell a team what its latency is.

Everything else in the ledger belongs to whoever builds the deck. Persona and tool categories are filing cuts that keep a deck from going out with five borrowing cards in a row. The tag columns — status, tradition, time horizon, vocabulary, sensitivity — calibrate a deck to a room, and the ledger's README defines them. None of it belongs on a card a player is holding.

Location Cards

Places in space where people live, work, or pass through. Locations are what create friction — distance from Earth, latency, scarcity, nobody agreeing whose law applies, physical danger.

Locations are organized by zone:

- Near Earth — close enough for real-time communication. Earth rules mostly apply. Examples: LEO Platforms, Lunar Bases, Earth Spaceports.
- Solar System — messages take minutes to hours. Local communities make their own rules. Examples: Mars Settlements, Asteroid Claims, Europa Outposts.
- Deep Space — messages take weeks to years. Communities handle everything alone. Examples: Generation Ships, Comet Claims, Probe Investments.

Question to ask of any location:

- How long does a message take to reach Earth?
- If something goes wrong, how long until help arrives?
- What's scarce here?
- Whose rules apply?

Persona Cards

People who need financial services. The persona answers the question:

“Whose problem are you actually solving?”

Persona categories — these are filing cuts for building a balanced deck, not something a team reasons about in play:

- Production — miners, farmers, technicians, scientists.
- Commerce — merchants, smugglers, entrepreneurs, prospectors, brokers.

- Services & Labor — doctors, contractors, migrant workers, ship crews.
- Capital & Authority — Earth investors, bankers, heirs, customs officers.
- Life Stages — children, retirees, tourists.

Note on Smugglers: the persona is a design provocation, not an endorsement. Frontier economies always include informal trade. If a session's context makes the framing awkward (compliance training, public-sector audiences), substitute "Informal Trader." This is an explicit option, not a workaround.

Financial Tool Cards

Tools that exist on Earth. The design challenge is figuring out how each would have to change to work in the location your team drew. Tool categories, again for deck-building rather than play:

- Borrowing — loans, bonds, mortgages, sukuk.
- Ownership — shares, venture capital, tokenized equity.
- Protection — insurance, futures, derivatives, parametric insurance, prediction markets.
- Transfer — remittances, letters of credit, escrow, trade credit, automated market makers.
- Time — savings, pensions, trusts, profit-sharing contracts, programmable escrow.
- Collective & Access — crowdfunding, mutual aid pools, grants, leasing, credit lines, rotating savings clubs, waqf.
- Governance — freeports, spectrum and orbital slots. These are legal structures rather than instruments. They change what a jurisdiction permits rather than how value moves.

The ledger marks every card Core or Variant, not just the tools. Core cards are the canonical deck; Variants are tested additions for particular contexts, and locations and personas have them as well. Print the Core set unless you have a reason not to. It is the general deck: 15 locations, 20 personas, 21 tools, enough for nine teams with spares. It also assumes Western finance vocabulary, which is its main limitation — the cards that break that assumption are waqf, sukuk, rotating savings clubs, mutual aid pools, profit-sharing contracts, religious endowments and cooperative stewards, all marked Variant. The five audience tabs are an extended menu of cards worth swapping in for particular rooms, not decks to print whole.

Adding Your Own Cards

The ledger is a working list, not a closed set. Sessions with a specific industry, region, or policy context often want a card the deck does not have.

A card earns its place by collision — the friction it creates against the other two draws. A card can be interesting on its own and still produce nothing in play.

- A location works by removing something a financial tool assumes: real-time communication, a reachable counterparty, an enforceable contract, a shared clock, somebody with jurisdiction.
- A persona works by having a need that the Earth-standard answer cannot meet once the location is applied.
- A tool works by carrying an assumption that visibly breaks somewhere in the deck. Name the assumption in the card's description. That is the line a team will use.

Test a new card by drawing it against three random cards from the other two types. If two of the three produce a problem you can state in one sentence, keep it. If none do, the card will read well and produce nothing.

New cards start as Variant. Promote one to Core after a session where it produced real work. If a Variant keeps producing dead draws, retire it.

The worked examples in Phase 1 show the same logic running the other way: the collision came first and generated the design.

How Card Drawing Works

One set of three cards per team, not per player. Teams draw together and design together.

A drawn card is out. It does not go back in the pile, so no two teams hold the same location, persona or tool. Print at least as many locations as you have teams, plus two or three spare — Phase 2 depends on locations being spread across zones rather than duplicated.

Redraws

Each team gets one redraw of one card in the first five minutes if they want it. The card they hand back goes to the bottom of its pile and can come out again; only cards a team keeps are out of play. After that, the cards are the cards. Teams that find their combination too easy or too hard should be pushed to design through it — weak collisions are part of what teams encounter in real frontier economies, and the facilitator has tools for this (see Facilitator Notes).

What the team does with the cards

Read all three. Spend two minutes on each, asking the questions in the canvas. Then look for the collision — the place where the location, persona, and tool create a problem that doesn't exist on Earth, or that exists on Earth in a different form. The collision is where the design lives.

Phase 1 — Design Your Concept

Each team has a place, a person, and a tool. The job is to invent a financial product, service, system, or institution that works for all three.

The Canvas

Five sections. Work through them in order. Don't skip the collision step — that's where the design comes from.

1. The Place

What makes this location difficult? How long do messages take? What's scarce? Whose rules apply?

2. The Person

What does this person need? What problems do they face that someone with the same role on Earth doesn't? Who pays them, and who do they pay?

3. The Tool

How does this tool work on Earth? What does it assume about communication, verification, enforcement, time? Which assumptions break in your location?

4. The Collision

What problems appear when you combine the three? Where is the friction? This is the most important step. If you can't name the collision in one or two sentences, you don't have a design problem yet — keep working.

5. Your Concept

What solves the problem? Give it a name and describe it in two sentences. Who benefits, who loses, what could go wrong, and what does it assume about its environment that could change?

Worked Example: Orbital Cycle Insurance

Cards: Europa Outposts + Migrant Workers + Insurance.

Place: Europa orbits Jupiter. Jupiter blocks comms with Earth on a predictable schedule, sometimes for hours. Radiation is high. Subsurface stations are isolated.

Person: A worker sending money home to family on Earth. If something happens to her, the family needs support. Verification of any incident is impossible from Earth.

Tool: Insurance traditionally requires investigation — an adjuster confirms what happened, courts enforce contested claims. Neither is feasible across this distance.

Collision: Verification is impossible in real time, but Europa's communication blackouts follow a regular orbital pattern.

Concept: Orbital Cycle Insurance. Payouts are triggered automatically by sensor data — station integrity, life support status, biometric signals — with payment going out during the

next Earth communication window. The orbit becomes the clock. The insurer never investigates; the sensors do.

Why this works as an example: it shows the pattern. The collision — nobody can verify a claim from Earth, but the blackouts run on a schedule — produced the design. Sensors fire the payout, and the money moves in the next communication window. Players who get stuck on “what would the concept be?” haven't spent enough time on collision.

Worked Example: Generational Futures

Cards: Comet Claims + Heirs + Futures.

Place: A comet carrying ice and organics passes through every 76 years. You can extract from it only when it's reachable.

Person: Inherited mining rights to the comet from a grandparent. The next pass is in 40 years. May be dead by then.

Tool: Futures contracts lock in prices for delivery later. Standard maturities are months, not lifetimes.

Collision: The asset is real but unreachable for decades. The contract has to outlive its parties.

Concept: Generational Futures. Tradeable contracts that span lifetimes. Losses are spread across everyone who has ever held the contract — a kind of intergenerational mutual aid attached to the asset itself. The contract is the asset; the comet is incidental.

Phase 2 — Build the Ecosystem

For Full Game and Two Sessions formats only. Quick Play stops after sharing.

Phase 2 takes the concepts from Phase 1 and asks how they connect. No financial system exists alone — banks need exchanges, insurers need other insurers to carry part of the risk, contracts need courts or something that does a court's job. The ecosystem phase makes that explicit.

Connection Tags

At the end of Phase 1, each team fills in four tags. These travel with the concept into Phase 2, and they are what you use to build clusters.

Two of the tags describe traffic — what moves in and out when the concept is working normally. Teams tick as many as apply, for both EXPORTS and IMPORTS.

- **Goods.** Ore, ice, fuel, food, parts. A mining claim exports ore and imports everything it cannot make on site.
- **Value.** Payment, credit, a claim on something, a share. Not necessarily currency — a remittance corridor moves value even when no money crosses the gap.
- **Information.** Prices, sensor readings, a verified record of what happened. Orbital Cycle Insurance exports the fact that a trigger fired.
- **People.** Crews, workers, specialists. A rotation camp exports trained labor and imports replacements.
- **Trust.** A record another team can check before dealing with a stranger. If your team built something that lets one party vouch for another, you export trust.
- **Something else.** Write it in. DeFi-style concepts often export composability — other protocols can build on them. Others export jurisdictional cover, or coordination. Don't force a fit.

The other two describe dependency — what has to exist for the concept to work at all, and what it can hand another team.

- **NEEDS.** Someone to verify a claim. A place to hold value while a deal completes. A rule about whose clock counts. A buyer at the far end.
- **OFFERS.** A regular moment when accounts get squared up. A record of who has failed to pay. Spare capacity. A set of rules other teams can operate under.

The difference matters when you form clusters. Exports and imports tell you who trades with whom. Needs and offers tell you who cannot function without whom.

The canvas gives the write-in a short line. Tell teams to keep it to a word or two and put any longer version on the back of the sheet.

Forming Clusters

Group teams into ecosystem clusters of 2–4 teams each. Three methods, depending on format:

In-person, by zone

Group teams whose locations share a zone. Asteroid belt teams together, Earth-orbit teams together, deep-space teams together. Simplest method but produces uneven clusters when zone draws are uneven.

In-person, by trade

Match teams whose tags complement each other. A team that exports goods pairs with one that imports them. A team that needs trust pairs with one that offers it. More effortful for the facilitator but produces tighter clusters.

Online, between sessions

The facilitator does this between Session 1 and Session 2. Collect canvases, identify clusters, brief participants 24 hours before Session 2 with their cluster mates' concepts.

Mapping the Cluster

Within each cluster, teams spend 15 minutes on three questions:

- Who pays your person, and who do they pay? Trace the flow.
- What flows between teams in this cluster? Goods, value, information, people, trust?
- Who has the most power? What single failure breaks the whole cluster?

Designing the Shared Institution

Each cluster identifies its biggest shared friction point and designs one shared institution to address it. The four canonical options:

Clearinghouse — settles trades between strangers. Stands between buyer and seller, and takes the loss if one of them fails to deliver. Earth examples: stock exchange clearing, futures markets.

Trust network — records reputation. Builds and shares information about who can be trusted to do what. Earth examples: credit bureaus, professional licensing, rating agencies.

Emergency fund — pools risk. Members contribute; the fund covers shocks too big for any one party. Earth examples: deposit insurance, mutual aid associations, sovereign reserve funds.

Governance structure — makes rules. Decides what counts as a valid contract, claim, or member. Earth examples: regulators, standards bodies, cooperative boards.

A cluster can also propose something outside these four — a regulatory body, a protocol, a jurisdiction. Use the four as starting points, not as a fence.

Worked Example: Belt Triangle

Three teams, all in the asteroid belt:

- Team A: Ore Futures Exchange — miners sell contracts for ore not yet extracted.
- Team B: Transit Bonds — ship captains issue bonds to finance long voyages, repaid from cargo proceeds.

- Team C: Hab Shares — those born on Ceres receive equity in the station's commercial economy.

Connection: Team A's miners need Team B's ships to move ore to market. Team B's bonds are backed by Team A's contracts. Team C's station shares appreciate when belt activity increases, giving residents a stake in both A and B.

Friction: Three teams pricing the same things differently, with no shared way to settle up. A failed contract from Team A breaks Team B's bonds and damages Team C's share value.

Shared institution: Belt Clearing Union. Sensor network verifies ore quality at extraction. Reputation system tracks promises kept and broken across all three instruments. Shared fund covers the failure of any one member, up to a limit.

Worked Example: Earth–Mars Bridge

Three teams across zones:

- Team A (Near Earth): Earth Spaceports + Customs Officers + Letters of Credit — a digital letter-of-credit system for cargo crossing into space.
- Team B (Solar System): Mars Settlements + Migrant Workers + Remittances — a remittance corridor for Mars workers sending earnings to Earth families.
- Team C (Solar System): Earth–Mars Cyclers + Ship Crews + Trusts — long-duration trusts for crew members whose families inherit while they're still in transit.

Connection: All three depend on a common record of identity that survives across the Earth–Mars latency. Team A needs to know cargo recipients are who they claim. Team B needs the worker's family to be reachable. Team C needs to track who has died and who has inherited while the crew was offline.

Shared institution: A federated identity registry maintained on both Earth and Mars, syncing during communication windows. Functions as the trust layer beneath all three financial flows.

Why this example matters: it crosses zones. Most clusters won't be neatly inside one. The Belt Triangle is the easy case; this one shows what most clusters actually look like.

Crisis Cards

Optional. Built into Quick Play + Crisis, and an extension to Full Game. After teams have shared their concepts, one Crisis Card is played to the whole room. Teams have 10 minutes to adapt.

The Card Ledger has eight, on their own sheet. They are not dealt into the three piles — you hold them. Choose the card in advance to fit what the teams have actually built, then reveal it as a draw. If every concept assumes reliable comms, play Relay Failure. If they all lean on one central party, play The Big Counterparty Lied. Blind drawing risks a crisis that touches nothing.

- Relay Failure — the link to Earth goes down for six months. Breaks verification and settlement.
- The Ship Doesn't Come — resupply misses its window. Breaks pricing and what counts as value.
- Earth Changes the Rules — a new law lands, and you hear four months late. Breaks jurisdiction, and assumes you knew in time.
- The Fast Route Opens — a two-year journey now takes four months. Breaks anything priced on distance.
- The Currency Slips — the Earth currency your contracts are written in halves in a week. Breaks store of value.
- The Big Counterparty Lied — the largest player has been misstating what it holds. Breaks reputation and trust networks.
- A Generation Arrives — population doubles, mostly people born there who signed nothing. Breaks membership and consent.
- Clocks Diverge — the parties no longer agree what time it is. Breaks maturity dates, interest, and claim windows.

Adaptation prompts

Hand teams these four questions when the Crisis Card is drawn:

- Can your system keep working? For how long?
- What breaks first?
- What's the smallest change that would let you survive this?
- What did your concept assume about the world that turned out not to hold?

The fourth question is the important one. Every Earth financial system has an assumption that broke at some point — inflation broke fixed-rate pensions, the internet broke physical settlement, climate is breaking parametric agriculture. The point of the Crisis phase is to surface those assumptions before they break in production.

Debriefing the Session

Every format ends in debrief. The two-line “what did we learn?” prompt produces polite generalities. Use these instead, calibrated to your audience.

Default debrief (any audience)

Three rounds, 5 minutes each, written first then shared:

- Which concept surprised you, and why?
- Which Earth assumption was hardest to drop?
- What would you take from today into your real work?

For practitioners and experts

Sharper version, for rooms that find the default too soft:

- Which assumption proved most expensive when challenged?
- Where did your concept quietly recreate something Earth already has?
- What's one thing you'd build differently now that the cards have shown you the friction?

For students and novices

More structured, for rooms still building the muscle:

- What did you assume about how money works that turned out to be more specific than you thought?
- Which Earth concept now makes more sense to you than it did this morning?
- What's a question you'd want to ask someone who actually works in finance?

For mandatory or low-engagement contexts

Don't fight the room. Acknowledge it. Two questions, written, no obligation to share:

- What was useful, even if just for you?
- What would have made it more useful?

Then end on time. A short, honest debrief beats a long one that asks people to perform engagement.

Facilitator Notes

This section addresses the things that go sideways in actual sessions. Read it before you run the game.

Reading the Room

The game's default framing assumes a curious, voluntary audience comfortable with speculative thinking. That's one room type. Match your opening to the actual room.

Voluntary, curious group

Use the default opener: "What happens to money when a message takes hours to arrive?" Lean into the speculative frame. This room signed up for it.

Mixed corporate / professional

Open with the historical anchor, not the speculative question. Something like: "Insurance, banking, contracts — every one of these was invented to solve a specific problem. Today we're going to reverse-engineer what it takes to invent one. The space framing is the prompt; the actual exercise is design." Gives the skeptical participants a real reason to engage.

Mandatory training, low engagement

Acknowledge the room. "I know some of you didn't pick this. Here's the deal: this exercise will produce two things — a couple of concepts you'll probably remember, and a clearer view of why the financial systems you use every day work the way they do. That's the offer." Then move briskly. Don't over-explain. Don't plead. Some people will engage anyway; that's the win.

Cross-cultural / mixed-fluency

Slow down on vocabulary in the first 10 minutes. Pull up the extended descriptions for any term that draws blank looks. Invite the room to surface their own examples — if you ask "how does this work where you're from?", you'll often get richer answers than the cards prompt.

Expert / finance-fluent

Pull the Expert & Fintech cards into the deck. Open with: "The default deck is calibrated for breadth of access. You will outgrow it. Treat the canvas as scaffolding, not a constraint, and add tools you think the deck is missing." Build in a short synthesis from you at the end — expert rooms want the facilitator's read on the patterns across the room.

Common Trouble Patterns

The team that's done in 20 minutes

Hand them a follow-up prompt without making it busywork. Two options:

- "What's the dark side of your concept? Who does it hurt?"
- "What does the regulator who hates this say?"

Both deepen the design without restarting it. Don't give them a second card draw — it telegraphs that finishing fast is wrong.

The team that's stuck on a soft collision

Some card combinations don't produce strong friction. Earth Spaceports + Tourists + Credit Lines is technically a valid draw with no obvious problem to solve. When you see it:

- Push them to invent something that exploits the easy conditions, not overcomes them. Buying cheap in one place to sell dear in another, stacking one set of rules on top of another, premium products that work because the friction is low.
- Or shift the question: "If this is the easy case, what makes it harder when the next location over isn't this easy?"

The disengaged or dominant participant

Three moves, in order of preference:

- Give them a job. Timekeeper. Scribe. Person who reports out at sharing. Lets them participate without being asked to perform.
- Pair them deliberately. Put a senior dominator with a strong second voice; put a disengaged participant with someone who will draw them out gently.
- Split the team. If a single team has stalled because of one person, you can break it into pairs for the design phase and reunite for sharing. Last resort, but real.

The team that performs irony

Some teams produce deliberately absurd concepts as a coping mechanism. They're engaged — just not in the register the game expects. Two moves:

- Lean in once. Let the absurd concept land at sharing. It'll get a laugh, and the room's energy will shift.
- Then redirect. Before the next team shares, name the move: "That was funny. Next we'll hear something different." You're not scolding the comedy team; you're protecting the next team's sincere work from being calibrated against irony.

The team that reinvents an Earth system

Especially common in cross-cultural sessions. A team builds a recognizable analog of an existing system — OFW remittance corridors, hawala, Islamic finance, ROSCA. Don't correct them. Ask: "What does space change about this version that wouldn't apply to the Earth version?" Often the team didn't know they were modeling something real, and the question makes the work generative again.

The team that won't play

It happens. A senior person who refuses the speculative frame, or a participant whose context makes engagement unsafe. Don't fight it. Note that they participated; let them deliver a short, low-effort share-out without comment; move on. The session's success doesn't depend on full engagement from every team.

Time Management

Three patterns to watch for:

- Sharing always overruns. Four minutes per team including the handover. Use a visible timer. Cut hard at the limit — questions can come at the end.
- Phase 1 finishes unevenly. Have a follow-up prompt ready for fast teams. Don't let slow teams panic — walk over and ask one focused question.
- Debrief gets squeezed. Protect it. If something has to give, let sharing run shorter, not the debrief. The debrief is where learning consolidates.

When the Premise Gets Pushback

Some participants reject the speculative frame. “Generation ships aren't realistic.” “We're not going to be on Mars in our lifetimes.” This is a real concern, especially with near-term industry audiences.

Two moves:

- Anchor closer. Pull them toward the Near Earth and Solar System cards — LEO Platforms, Lunar Bases, Asteroid Claims. The speculation horizon shortens. The financial questions stay live.
- Reframe the exercise. The point isn't prediction; it's reasoning. “Whether or not we're on Mars in 30 years, the design problem is the one finance has solved many times before. We're practicing the reasoning.”

If a participant continues to reject the frame, see “the team that won't play” above.

After the Session

What happens after often matters more than what happens during, especially in organizational contexts. Two small affordances:

- Send participants away with a sentence they can use if asked. Something like: “We did a design exercise about how financial systems get invented when the usual ones don't reach.” Saves them having to invent a description for skeptical colleagues.
- If the organizer asks how it went, give an honest answer. Not every session is a triumph. A 30% engagement session that produced two real concepts is a partial win and worth saying so.

A successful session isn't one where everyone was happy. It's one where the people who engaged learned something, and the people who didn't weren't actively turned off the topic.

Materials

Physical session

- Card deck, printed from the Card Ledger. Core set by default.
- Canvas worksheets, one per team. Template in the appendix.
- Large paper or wall space for ecosystem mapping (Phase 2).
- Markers, sticky notes, timer.

Online session

- Video call with breakout rooms.
- Shared whiteboard (Miro, Mural, FigJam, or equivalent) with the canvas pre-loaded for each team.
- Card draws can be done by sharing the relevant tab of the Card Ledger and assigning each team a row range — or by the facilitator drawing on participants' behalf.
- Ecosystem clustering uses a separate whiteboard frame with team boxes that can be moved into clusters.

Producing the deck

The ledger is a spreadsheet, not a print file. To get a physical deck, filter it to the cards you want and lay them out at a size a team can pass around. A6 or standard playing-card size both work.

On the card face:

- The name.
- The short description.
- Which of the three types it is, carried by color or by the card back rather than a printed word.
- On a location card, the zone and its Latency line.

Off it: category, status, tradition, time horizon, vocabulary, sensitivity, and the facilitator notes. The extended description is a judgment call. Print it if you want the detail available without a team having to ask; leave it off if you would rather they ask.

The zone alone is not enough. It tells a team roughly how far out they are; it does not tell them the number they design against. Print the latency with it — “1.3 seconds each way,” “3 to 22 minutes each way,” “none perceptible” — The ledger carries it in its own Latency column; print that line beneath the zone.

Print one deck for the session, not one per team. Teams draw from the same three piles. Aim for 15–20 cards per type — enough variety that draws feel distinct, few enough that they don't feel random.

An online session needs no printing. Share the relevant ledger rows with each team, or draw on their behalf and post the three cards into their breakout room.

Who Is This For?

Three audience types, broadly. The Core set works for all of them; the ledger's audience tabs suggest what to add or drop.

Education

Business, engineering, policy, and design students learning how systems develop under constraints. The game scales down well; for school-age groups keep to the Near Earth and Solar System locations and leave the expert tools out of the deck.

Industry

Space industry, financial services, fintech, insurance, foresight teams. Useful for teams thinking about extreme constraints, frontier markets, or the underlying logic of financial systems they work with daily.

Everyone else

Science fiction readers, design researchers, anyone curious about how money and trust actually work. The game doesn't require finance background — the Core set is calibrated for that.

A Note from the Designers

The point isn't to predict space economics. The point is to develop the muscle for thinking about how financial systems get built when the usual rails don't reach.

That muscle transfers. Frontier markets, post-conflict reconstruction, climate adaptation, AI-mediated economies, crypto-native systems — all of them require the same kind of reasoning the game asks for. The space framing is the prompt. The reasoning is the point.

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Changeist runs Money for Mars as a facilitated session, half-day or ninety minutes, and builds custom decks for a particular sector or region. We also train facilitators to run it in-house. Start at moneyformars.com.

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Scott Smith and Susan Cox–Smith

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github.com/Changeist/money-for-mars

Appendix — Terms on the Cards

Some cards name real financial institutions that a room may not know. One line each, enough to keep a team moving. Several are older and more widespread than their Western equivalents.

- Hawala — a trust-based money transfer network with no money actually crossing the border. A broker at each end settles between themselves later. A hawaladar is one of those brokers.
- OFW — Overseas Filipino Worker. The Philippines' migrant labor corridor, and one of the largest and best-documented remittance systems in the world.
- ROSCA — rotating savings and credit association. Members pay in regularly and each takes the whole pot in turn. Known as arisan in Indonesia, hui in China, susu in West Africa and the Caribbean, tanda in Mexico.
- Waqf — a permanent endowment in Islamic law. The principal is locked and cannot be sold or inherited; the income funds a stated public purpose, sometimes for centuries.
- Takaful — cooperative insurance. Members contribute to a common pool and claims are paid from it, so members are both the insurer and the insured.
- Mudaraba and musharaka — profit-sharing investment contracts. In mudaraba one party provides capital and the other the work; in musharaka both contribute capital. Returns are a share of profit rather than interest.
- Sukuk — a certificate giving the holder a share in a real underlying asset and the income it produces, rather than a debt paying interest.
- Flag state — the country whose law a ship or spacecraft operates under, which need not be where its owners or crew are from. The reason a vessel's legal home can be chosen rather than inherited.
- Freeport — a zone where goods can be stored, traded and transferred without triggering the surrounding jurisdiction's taxes or tariffs. Value sits between legal systems rather than inside one.
- Anchor tenant — the large, usually government, customer whose committed spending makes a piece of infrastructure financeable. The design question is what happens when they leave.
- Order book — the running list of buy and sell offers on an exchange. Matching a trade means finding a counterparty in it, which is what an automated market maker does away with.
- Parametric insurance — a policy that pays out when a measurable trigger fires, with no claim and no investigation. Already used in space operations and in climate cover.
- DAO — decentralized autonomous organization. A group whose membership rules and treasury are enforced by software rather than by a charter and a board.

Appendix — The Canvas

One per team. Print at A3, landscape. The file is in the repository at github.com/Changeist/money-for-mars; the version below is for reference, not for use — it will not give a team enough room to write at this size.

For an online session, load the same layout onto a whiteboard frame per team instead of printing it.

MONEY FOR MARS THE CANVAS

TEAM _____ DATE _____

LOCATION DRAWN

PERSONA DRAWN

TOOL DRAWN

ZONE _____ LATENCY _____

1 THE PLACE

What makes this location difficult? How long do messages take? What is scarce? Whose rules apply?

2 THE PERSON

What does this person need? What problems do they face that the same role on Earth does not? Who pays them, and who do they pay?

3 THE TOOL

How does this work on Earth? What does it assume about communication, verification, enforcement, time? Which assumptions break here?

4 THE COLLISION

Where do the three cards create a problem that does not exist on Earth? If you cannot name it in two sentences, you do not have a design problem yet — keep working.

5 YOUR CONCEPT

NAME IT _____

Describe it in two sentences.

WHO BENEFITS

WHO LOSES

WHAT COULD GO WRONG

WHAT IT ASSUMES THAT COULD CHANGE

CONNECTION TAGS FILL AT THE END OF PHASE 1 — FULL GAME AND TWO SESSIONS ONLY

EXPORTS ☐ goods ☐ value ☐ information ☐ people ☐ trust ☐ other _____

NEEDS _____

IMPORTS ☐ goods ☐ value ☐ information ☐ people ☐ trust ☐ other _____

OFFERS _____

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